

Choice of step size

To get a solution with acceptable accuracy, we need to choose a step size that gives:

- a sufficiently small local error
$$h \leq h_{acc}$$

- absolute stability: $h \leq h_{stab}$

We say a problem is stiff if,
with an explicit method,

$$h_{stab} \ll h_{acc}.$$

Example: Prothero-Robinson

$$u'(t) = \lambda(u - \cos(t)) - \sin(t) = f(u, t)$$

with $\lambda < 0$.

If $u(t_0) = \cos(t_0)$, then the solution is simply $u(t) = \cos(t)$.

How does $u(t)$ behave for other initial data?

From Duhamel's principle,

$$u(t) = \cos(t) + e^{\lambda(t-t_0)}(u_0 - \cos(t_0))$$

What is h_{stab} ? $\mathcal{O}(1/\lambda)$

What is h_{acc} ? $\mathcal{O}(1/\lambda)$ initially
 $\mathcal{O}(1)$ later

(notebook)

Example: The Heat Equation

$$u_t = u_{xx}$$

$$u(x, t_0) = u^0(x) \quad u(0, t) = u(1, t) = 0$$

Exact solution: we use the ansatz

$$u(x, t) = \sum_{j=1}^{\infty} \hat{u}_j(t) \sin(j\pi x)$$

where $\hat{u}(0)$ is the Fourier transform of $u^0(x)$. Plug it in and equate coefficients:

$$\hat{u}'_j(t) = -j^2 \pi^2 \hat{u}_j(t)$$

so
$$\hat{u}_j(t) = e^{-j^2 \pi^2 t} \hat{u}_j(0)$$

and

$$u(x, t) = \sum_{j=1}^{\infty} \hat{u}_j(0) e^{-j^2 \pi^2 t} \sin(j\pi x)$$

(show notebook and discuss)

Semi-Discretization of the heat equation

$$U_j'(t) = \frac{1}{(\Delta x)^2} (U_{j+1} - 2U_j + U_{j-1})$$

$$U'(t) = \frac{1}{(\Delta x)^2} \underbrace{\begin{bmatrix} -2 & 1 & & \\ & 1 & -2 & \\ & & \ddots & \ddots \\ & & & 1 & -2 \\ & & & & 1 & -2 \end{bmatrix}}_A U(t)$$

$$\Rightarrow U(t) = e^{tA} U(0)$$

$$A = R \Lambda R^{-1}$$

$$U(t) = R e^{t\Lambda} R^{-1} U(0)$$

$$\hat{U} = R^{-1} U$$

$$U(t) = R e^{t\Lambda} \hat{U}(0)$$

It turns out that

$$R_{ij} = \sin(j\pi x_i) \quad \Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_m)$$

So we have

$$U_i(t) = \sum_{j=1}^{\infty} \hat{U}_j(0) e^{\lambda_j t} \sin(j\pi x_i)$$

One can show that

$$\lambda_j = \frac{2}{\Delta x^2} (\cos(j\pi \Delta x) - 1)$$

so

$$-\frac{4}{(\Delta x)^2} < \lambda_j < -\pi^2$$

We again have a hierarchy of rates of change, with exponential decay to slower and slower manifolds.

After some short time, the local error is independent of the fast dynamics, but they always affect the stability:

$$h_{stab} = \mathcal{O}(1/\Delta x^2)$$

For stiff problems, explicit methods are inefficient. They force us to compute an answer that is much too accurate!

Instead, we want to use a method such that $\lambda h_{acc} \in S$ even when $|\lambda| \gg 1$.

How large is h_{acc} ? Remember that LTE is proportional to some derivative of the exact solution. A reasonable first approximation is

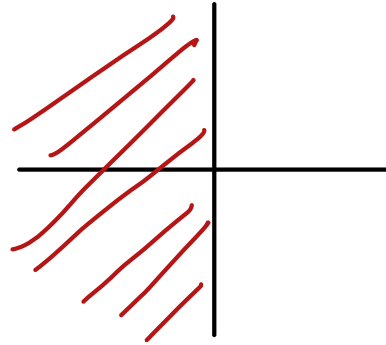
$$h_{acc} \sim \frac{1}{\|u'(t)\|}.$$

For PR or the heat equation, since λ is real and negative, we just need a method that's stable on the negative real axis.

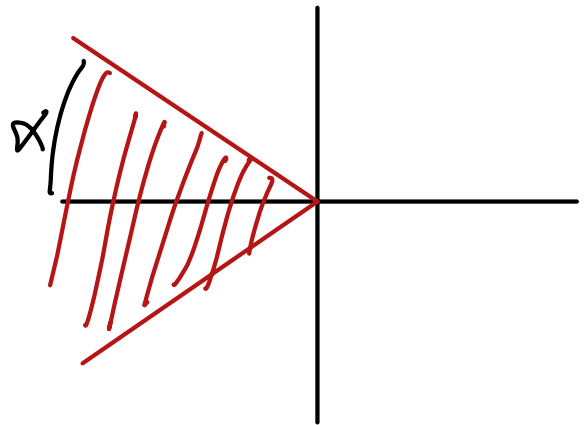
More generally, we look for methods that are stable for any stable ODE:

$\zeta \in S$

A-stable



$A(\alpha)$ -stable:



L-stability

For extremely stiff problems, like the heat equation when $\Delta x \rightarrow 0$, we should consider what happens when $h\lambda \rightarrow -\infty$. The true solution in this case decays extremely fast. The numerical solution will do the same if

$$\lim_{z \rightarrow -\infty} |R(z)| = 0 \quad (1)$$

If a method is A-stable and satisfies (1), we call it L-stable.

(notebook)

Homework

Implement the 5-stage diagonally implicit RK Method from HW vol. 2, section IV.6, Table 6.5 ^(on p. 100) (just use the first row of weights - you don't need to implement the error estimator). Apply it to the problem

$$u'(t) = \lambda(u - \phi(t)) + \phi'(t)$$

with

$$\phi(t) = e^{-t} \sin(10t) + \cos(20t)$$

and $\lambda = -10^4$.

Solve up to $T=10$ and plot the error at the final time versus h for $h \in [10^{-6}, 10^{-1}]$ (on a log-log scale).